Software Implementation

Agenda

- Software implementation definition
- key events during software implementation
- Software changeover methods

Implementation

- Implementation means putting something into effect.
- Software implementation is the installation of the new system on end users site.
- The key events in this stage are:
  i. *Data conversion*
  ii. *System change-over:*
  iii. *Staff training*

Implementation

- The key events in this stage are:
  i. **Data conversion:** This is the process of converting data stored in files on the old system into correct format in the new system.
  ii. **System change-over**: It is the is the process of putting the new information system on line and retiring the old system. It can also be defined as the process of switching on the new system and switching off the old system.
  iii. **Staff training:** This is the process of showing/familiarizing the end user of the system on how to use the system. User documentation is often used during training session.

Change-over methods

There are four methods of system change-over:

i. **Direct cutover**:

- The old system is stopped completely and the new system is started.
- All the data that used to be input into the old system now goes into the new one.

Change-over methods - **Direct cutover**

Advantages:

- It is the least expensive method
- It takes minimum time and effort
- The new system is up & running immediately

Change-over methods - **Direct cutover**

Disadvantage:

- It involves more risks than other changeover methods in that if the new system fails, there is no backup system so data can be lost.

Change-over methods

ii. **Parallel operation**:

- The parallel operation changeover method requires that both the old and the new information systems operate fully for a specified period.
- Data is input to both systems and output generated by the new system is compared with the equivalent output from the old system.
- When users, management, and IT group are satisfied that the new system operates correctly then the old system is terminated.

Change-over methods - **Parallel operation**

Advantages:

- It involves lower risks.
- If the new system fails, the old acts as a backup system.
- The outputs from old and new system can be compared to check that the new system is running correctly.

Change-over methods - **Parallel operation**

Disadvantages:

- It is the most costly changeover method.
- Running and entering data into two systems takes a lot of extra time and effort.

Change-over methods

iii. Pilot operation:

- The pilot changeover method involves implementing the complete new system at a selected location of a company (e.g. an office or a department).
- The group that uses the new system first is called the ***pilot site***.

Change-over methods - **Pilot operation**

Advantages:

- All features of the system can be fully trialled.
- If the new system fails only small part or section of an organization is affected.
- The staff who are part of pilot scheme can help train other staff.

Change-over methods - **Pilot operation**

Disadvantage:

- If the new system fails, the pilot site is affected since there is no backup system.

Change-over methods

iv.Phased operation:

- The phased operation changeover method involves implementing the new system in stages, or steps gradually replacing the old system until eventually the new system has taken over.

Change-over methods - **Phased operation**

Advantages:

- In this approach risk of errors or failures is limited to the implemented module only as well as it is less expensive than the full parallel operation.
- It allows users to gradually get used to the new system.
- Staff training can be done in stages.

Change-over methods - **Phased operation**

Disadvantage:

- If the new system fails, there is no backup system so data can be lost.